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The algorithm and complexity of co-secure domination in geometric intersection graphs

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Abstract
Given a graph G with vertex set V, a dominating set $S \subseteq V$ is a co-secure dominating set of G if for every vertex $u \in S$, there is a vertex $v \in V \setminus S$ adjacent to u such that $(S \setminus \{u\}) \cup \{v\}$ is a dominating set of G. The minimum co-secure dominating set (or, for short, MCSDS) problem asks to find an MCSDS in a given graph. In this paper, first we show that the decision version of the problem is NP-complete in grid graphs and supergrid graphs. Consequently, we show that the problem remains NP-complete for unit disk and unit square graphs. Secondly, we show that the MCSDS problem is APX-hard in d-box graphs for any fixed integer $d \geq 2$. Finally, we give an $O(n+m)$ time $2(t-1)$-approximation algorithm for the MCSDS problem in several geometric intersection graphs which are $K_{1,t}$-free for some integer $t \geq 3$.

Keywords Co-secure domination · Geometric intersection graphs · NP-complete · APX-hard · Approximation algorithm

Mathematics Subject Classification 05C69 · 68Q25

1 Introduction

Let $G = (V, E)$ be a simple, undirected graph with vertex set V and edge set E. Denote the *open* and *closed neighborhood* of a vertex $v \in V$ by $N_G(v)$ and $N_G[v]$, respectively, that is, $N_G(v) = \{x : vx \in E\}$ and $N_G[v] = N_G(v) \cup \{v\}$. A set $D \subseteq V$ is a *dominating set* (or, for short, *DS*) of G if for each vertex $u \in V \setminus D$, there is a vertex $v \in D \cap N_G(u)$. The *domination number* $\gamma(G)$ is the minimum cardinality among all DSs of G. A DS $S \subseteq V$ is a *co-secure dominating set* (or, for short, *CSDS*) of G if for every vertex $u \in S$, there is a vertex $v \in (V \setminus S) \cap N_G(u)$ such that $(S \setminus \{u\}) \cup \{v\}$ is a DS of G. In this case, we call v *replaces* u or v is a *replaceable vertex* of u. The *co-secure domination number* $\gamma_{cs}(G)$ is the

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minimum cardinality among all CSDSs of G. Note that if G has an isolated vertex, then there is no co-secure dominating set for G. For a graph without an isolated vertex, a maximum independent set is a co-secure dominating set (Arumugam et al. 2014). All graphs considered in the paper have no isolated vertices.

Co-secure domination was introduced by Arumugam et al. (2014). They proved that the decision version of the MCSDS problem is NP-complete in bipartite, chordal, and planar graphs, where the *MCSDS problem* is the problem of finding a minimum CSDS in a given graph. Kusum and Pandey (2023) showed that the problem remains NP-complete for split graphs. Zou et al. (2022) and Kusum and Pandey (2023) proposed linear-time algorithms for the MCSDS problem in proper interval graphs and cographs, respectively. The approximation aspects of the MCSDS problem have also been studied. Kusum and Pandey (2023) showed that the problem can be approximated within an approximation ratio of $(\Delta + 1)$ for perfect graphs with maximum degree Δ. Panda et al. (2024) showed that the problem can be approximated within an approximation ratio of $O(\ln |V|)$ for any graph G with $\delta(G) \geq 2$ and for 3-regular and 4-regular graphs they showed that the problem is approximable within a factor of $\frac{8}{3}$ and $\frac{10}{3}$, respectively. For the inapproximability of the MCSDS problem, Kusum and Pandey (2023) showed that the problem cannot be approximated within an approximation ratio of $(1 - \varepsilon) \ln |V|$ for perfect elimination bipartite graphs and star convex bipartite graphs unless P=NP and the problem is APX-complete for 3-regular graphs.

A graph $G = (V, E)$ is a *geometric intersection graph* if there is a bijection between V and a set P of geometric objects such that $uv \in E$ for two vertices u and v if and only if their corresponding objects intersect. The set P is called a *geometric intersection representation* of G. A d*-box* is a subset of $\mathbb{R}^d$ that is a cartesian product of d intervals in $\mathbb{R}$. A d*-box graph* is a geometric intersection graph when P is a set of d-boxes. Clearly, a 2-box is a rectangle and a 2-box graph is also called a *rectangle graph*. A rectangle graph is *proper* if no rectangle is properly contained within another. A *unit-height rectangle graph* is a geometric intersection graph when P is a set of unit-height rectangles (the height and width of all rectangles are 1 and at least 1, respectively). A *unit square graph* is a geometric intersection graph when P is a set of unit squares (the height and width of all squares are 1). When P is a set of unit disks (the diameters of all disks are 1), the corresponding geometric intersection graph is called a *unit disk graph*. When P is a set of unit balls (the diameters of all balls are 1), the corresponding geometric intersection graph is called a *unit ball graph*. If the centers of all disks of a unit disk graph have integer coordinates, then the graph is called a *grid graph*. Let S^∞ be an infinite graph such that its vertex set $V(S^\infty)$ contains all points of the Euclidean plane with integer coordinates and its edge set $E(S^\infty) = \{(u, v) : |u_x - v_x| \leq 1 \text{ and } |u_y - v_y| \leq 1\}$, where u_x and u_y represent the x and y coordinates of vertex u, respectively. A *supergrid graph* is a finite and vertex-induced subgraph of S^∞. In general, a supergrid graph is not a grid graph, and vice versa. In this paper, we will show that supergrid graphs are a subclass of unit square graphs.

Geometric intersection graphs have a large number of applications in practical life. For example, unit disk graphs are usually used to represent the geometrical structures of wireless sensor networks (assuming that all sensors have an equal transmission reception range). In a mountainous area or in an underwater terrain, the surface is often not flat. In such scenarios, wireless sensor network represents a unit ball graph. Rectangle graphs are widely applied in map labeling, VLSI design, and data mining. Supergrid graphs have been proposed for computing the stitch traces for computerized embroidery machines (Hung et al. 2015).

For co-secure domination, consider a network with sensors deployed at some vertices so that they can collectively cover the network. If changing conditions at a vertex, for example adverse weather, electronic interference, or denial of service, etc., affect the proper operation of the sensor, then consider an alternate location, usually nearby for this sensor and yet preserves the property that the sensors collectively cover the network.

The paper is organized as follows. In Sect. 2, we provide some preliminary results that will be used for future discussions. In Sect. 3, we show that the decision version of the MCSDS problem is NP-complete in grid graphs and supergrid graphs. In Sect. 4, we show that the MCSDS problem is APX-hard in d-box graphs for any fixed integer $d \geq 2$. In Sect. 5, we give an $O(n+m)$ time $2(t-1)$-approximation algorithm for the MCSDS problem in $K_{1,t}$-free graphs for some integer $t \geq 3$.

2 Preliminaries

In this section, we give further definitions and some useful results that will be used for future discussions. We refer to West (2001) for graph theoretic definitions and notations. Denote a path of order n by P_n. Given a graph $G = (V, E)$, a vertex $u \in V$ is called a *leaf vertex* in G if $|N_G(u)| = 1$. If there is a non-leaf vertex $v \in V$ joining a $P_2 = xy$, that is, $N_G(y) = \{x\}$ and $N_G(x) = \{v, y\}$, then v is called a P_2-*vertex*, x is called a P_2-*support vertex* and y is called a P_2-*leaf vertex*. Clearly, y is a leaf vertex. Let L_2 be the set of all P_2-leaf vertices in G.

Lemma 2.1 *Let $G = (V, E)$ be a graph. Then each CSDS S of G can be transformed in polynomial time to a CSDS S' of G such that $S' \cap L_2 = \emptyset$ and $|S'| \leq |S|$.*

Proof If $S \cap L_2 = \emptyset$, then let $S' = S$. Otherwise for each $y \in S \cap L_2$, we do the following operations. Let x be the unique neighbor of y and v be another neighbor of x other than y. Then $x \notin S$. If $v \notin S$ or $|N_G(v) \cap (V - S)| \geq 2$, then let $S' = (S \backslash \{y\}) \cup \{x\}$. Otherwise $v \in S$ and $N_G(v) \cap (V - S) = \{x\}$, then let $S' = (S \backslash \{v, y\}) \cup \{x\}$. Since v is not a leaf, $|N_G(v) \cap S'| \geq 2$. Thus in both cases, $|S'| \leq |S|$ and S' is a CSDS of G. Since all vertices $y \in S \cap L_2$ are considered, $S' \cap L_2 = \emptyset$. $\square$

For a graph $G = (V, E)$ and an integer $s \geq 0$, an s-*subdivision* of an edge $uv \in E$ is defined as a replacement of uv by a path with endvertices u and v and with s new internal vertices. Let $V = \{v_1, v_2, \ldots, v_n\}$ and $s_{ij} \geq 0$ be an integer for each edge $v_i v_j \in E$. Let G be a graph. Construct the graph G' from G by an s_{ij}-subdivision of each edge $v_i v_j \in E$ and joining a $P_2 = x_i y_i$ for each vertex $v_i \in V$, where $v_i x_i, x_i y_i \in E'$. For convenience, we call G' is a P_2-*suspending s_{ij}-subdivision graph* of G.

Lemma 2.2 *Given a graph $G = (V = \{v_1, v_2, \ldots, v_n\}, E)$ and an integer $s_{ij} \geq 1$ for each edge $v_i v_j \in E$. Let $G' = (V', E')$ be a P_2-suspending $5s_{ij}$-subdivision graph of G. Then*

(1) $\gamma_{cs}(G') = \gamma(G) + n + 2\sum_{v_i v_j \in E} s_{ij}$;
(2) every DS D of G can be transformed in polynomial time to a CSDS S of G' such that $|S| \leq |D| + n + 2\sum_{v_i v_j \in E} s_{ij}$;
(3) every CSDS S of G' can be transformed in polynomial time to a DS D of G such that $|S| \geq |D| + n + 2\sum_{v_i v_j \in E} s_{ij}$.

Proof Let $s = \sum_{v_i v_j \in E} s_{ij}$. For each edge $v_i v_j$ in G, the $5s_{ij}$-subdivision of the edge corresponds a path, denoted by $v_i q_1 \cdots q_{5s_{ij}} v_j$, in G'. Let $Q_{ij} = \{q_1, \ldots, q_{5s_{ij}}\}$. For each vertex $v_i \in V$, the path P_2 connected by v_i is denoted by $P_2 = x_i y_i$, where $v_i x_i, x_i y_i \in E'$, in G'.

Given a DS D of G. We will construct a CSDS S of G'. Firstly, add D and all x_i for $1 \leq i \leq n$ into a set S'. Secondly, for each edge $v_i v_j$ in G, if $v_i, v_j \notin D$, then add q_{5t-3}, q_{5t-1} for all $1 \leq t \leq s_{ij}$ into S'. Otherwise $v_i \in D$ or $v_j \in D$, without loss of generality, assume $v_i \in D$. Then add q_{5t-3}, q_{5t} for all $1 \leq t \leq s_{ij}$ into S'. Thirdly, delete v_i from S' if $N_{G'}[v_i] \subseteq S'$ for each $v_i \in V$. Let S be the updated set, that is, $S = S' \backslash \{v_i \in V : N_{G'}[v_i] \subseteq S'\}$. Clearly, $|S| \leq |S'| = |D| + n + 2s$. Now we show that S is a CSDS of G'. Clearly, S is a DS of G'. For each x_i, since there is at least one neighbor of v_i other than x_i in S if $v_i \notin S$, y_i replaces x_i. For each edge $v_i v_j$ in G, $S \cap Q_{ij}$ is a DS of $G'[Q_{ij}]$. Then for each $v_i \in S$, there is a neighbor q of v_i not in S by the construction of S, q replaces v_i. For each edge $v_i v_j \in E$, we consider the path $v_i q_1 \cdots q_{5s_{ij}} v_j$. For the case of $v_i, v_j \notin D$, q_{5t-4} replaces q_{5t-3} and q_{5t} replaces q_{5t-1} for all $1 \leq t \leq s_{ij}$. For the case of $v_i \in D$ or $v_j \in D$, we again assume $v_i \in D$ and further $q_{5t-3}, q_{5t} \in S$ for all $1 \leq t \leq s_{ij}$. Then q_{5t-2} replaces q_{5t-3} and q_{5t-1} replaces q_{5t} for all $1 \leq t \leq s_{ij}$. Hence, S is a CSDS of G'. Thus (2) holds. When D is a minimum DS of G, $\gamma_{cs}(G') \leq |S| \leq |D| + n + 2s = \gamma(G) + n + 2s$.

Given a CSDS S of G'. We will construct a CSDS S' of G' from S satisfying the following conditions: (1) $|S'| \leq |S|$; (2) for each $1 \leq i \leq n$, $y_i \notin S'$; (3) for each path $v_i q_1 \cdots q_{5s_{ij}} v_j$ and each $1 \leq t \leq s_{ij}$, $|S' \cap \{q_{5t-4}, \ldots, q_{5t}\}| = 2$.

By Lemma 2.1, S can be transformed in polynomial time to a CSDS S_1 of G' such that $|S_1| \leq |S|$ and $y_i \notin S_1$ for each $1 \leq i \leq n$. Then $x_i \in S_1$ for each $1 \leq i \leq n$. For each path $v_i q_1 \cdots q_{5s_{ij}} v_j$ and each $1 \leq t \leq s_{ij}$, we now show that $|S_1 \cap \{q_{5t-4}, \ldots, q_{5t}\}| \geq 2$. Clearly, $|S_1 \cap \{q_{5t-4}, \ldots, q_{5t}\}| \geq 1$. Suppose to the contrary that $|S_1 \cap \{q_{5t-4}, \ldots, q_{5t}\}| = 1$ for some $1 \leq t \leq s_{ij}$. Then $q_{5t-2} \in S_1$. Thus q_{5t-2} has no replaceable vertex, a contradiction.

For each path $v_i q_1 \cdots q_{5s_{ij}} v_j$ and each $1 \leq t \leq s_{ij}$ satisfying $|S_1 \cap \{q_{5t-4}, \ldots, q_{5t}\}| \geq 3$, if $v_i, v_j \in S_1$, then construct $S_2 = (S_1 \backslash Q_{ij}) \cup (\cup_{t=1}^{s_{ij}} \{q_{5t-3}, q_{5t-1}\})$. Otherwise $v_i \notin S_1$ or $v_j \notin S_1$, without loss of generality, assume $v_i \notin S_1$, then construct $S_2 = (S_1 \backslash Q_{ij}) \cup (\cup_{t=1}^{s_{ij}} \{q_{5t-3}, q_{5t}\} \cup \{v_i\})$. Further if $N_{G'}[v_j] \subseteq S_1$, then let $S' = S_2 \backslash \{v_j\}$. In either case, S' is a CSDS of G' and satisfies the conditions (1), (2) and (3).

Let $D = V \cap S'$. In the following we show that D is a DS of G. If $v_j \notin D$, then $v_j \notin S'$. By condition (2), y_j is unique replaceable vertex of x_j. Then there is $q_{5s_{ij}} \in S'$, where $q_{5s_{ij}}$ is on the path $v_i q_1 \cdots q_{5s_{ij}} v_j$. Combined with condition (3), either $\{q_{5t-4}, \ldots, q_{5t}\} \cap S' = \{q_{5t-3}, q_{5t}\}$ or $\{q_{5t-4}, \ldots, q_{5t}\} \cap S' = \{q_{5t-2}, q_{5t}\}$ for any $1 \leq t \leq s_{ij}$. In either case, $q_1 \notin S'$, then $v_i \in S'$, otherwise q_2 has no replaceable vertex or q_1 is not dominated by S'. Then $v_i \in D$, it follows that v_j is dominated by v_i in D. Thus D is a DS of G. By conditions (1), (2) and (3), $|S| \geq |S'| = |D| + n + 2s$. Then (3) holds. When S is a minimum CSDS of G', $\gamma_{cs}(G') = |S| \geq |D| + n + 2s \geq \gamma(G) + n + 2s$. Combined with above inequality, $\gamma_{cs}(G') = \gamma(G) + n + 2s$, it follows that (1) holds. $\square$

3 NP-completeness

In this section, we show that the decision version of the MCSDS problem is NP-complete on grid graphs and supergrid graphs by giving a polynomial-time reduction from the minimum dominating set problem which is known to be NP-complete on planar graphs with maximum degree 3 (Garey and Johnson 1979). The associated decision problems are defined below.

Co-Secure Domination in graphs (CSDM)
Instance: A graph G, $j \in \mathbb{N}^+$.
Question: Does G have a CSDS of size at most j?
Domination in Planar graphs (DM-PLA)

Instance: A simple planar graph G with maximum degree 3, $k \in \mathbb{N}^+$.

Question: Does G have a DS of size at most k?

Let G be a planar graph. An *orthogonal drawing* of G is an embedding of G in the plane in such a way that its vertices are located at points with integer coordinates and its edges are drawn as a sequence of connected line segments of the form $x = i$ (vertical) or $y = j$ (horizontal) for integers i and j. A graph G admits an orthogonal drawing if and only if it is planar with maximum degree at most 4 (Valiant 1981).

Lemma 3.1 (Biedl and Kant 1998) *For a planar graph G with vertex set V and maximum degree at most 4, there is a linear time algorithm that produces an orthogonal drawing of G with, along each edge e, at most 2 bends (i.e., a point of e where a horizontal and a vertical segment meet) and using $O(|V|^2)$ area.*

Theorem 3.2 *CSDM is NP-complete in grid graphs.*

Proof For a given graph $G = (V, E)$, a set $S \subseteq V$ and $j \in \mathbb{N}^+$, we can verify whether S is a CSDS of G with $|S| \le j$ or not in polynomial time. So CSDM is in NP. In what follows we offer a polynomial time reduction from the NP-complete DM-PLA problem. Let $G = (V, E)$ be a simple planar graph with maximum degree 3, where $V = \{v_1, v_2, \ldots, v_n\}$ and $|E| = m$. We use the similarly way as in Wang et al. (2023b) to construct a grid graph $G' = (V', E')$ from G. In the following we describe the construction:

Step 1. First construct an orthogonal drawing of G by Lemma 3.1 (see Fig. 1b). Secondly, expand the drawing by 10 times, denoted the resulted drawing by Γ', and for each vertex v_i $(1 \le i \le n)$, label still as v_i the grid point at which v_i is located in Γ'. Obviously, Γ' is also an orthogonal drawing of G. Call the sequence of connected line segment(s) in Γ' representing the edge $v_i v_j$ is the $v_i v_j$-*line segment sequence*. Obviously, the length of the $v_i v_j$-line segment sequence is divided by 10. We divide the sequence into l_{ij} parts with each being of length 10.

Step 2. Since maximum degree of G is 3, there is a vertical or horizontal direction around each grid point labelled as v_i which is blank in Γ'. On the direction, we add a straight segment S_i of length two with an endpoint v_i and label the grid points as x_i, y_i in turn (see example x_1, y_1 in Fig. 1e), denoted the resulted drawing by Γ''.

Step 3. For each of $v_i v_j$-line segment sequence, replace the *part* P (which is a line segment of length 10 that containing v_i or v_j but not both (except the case $l_{ij} = 1$) on the $v_i v_j$-line segment sequence) with the shape as presented in Fig. 1c if P is horizontal, otherwise with the shape as presented in Fig. 1d, denoted the resulted sequence by L_{ij}. Note that the length of L_{ij} is $10 l_{ij} + 6$. Denote the resulted drawing by Γ (see Fig. 1e) and $l = \sum_{v_i v_j \in E} l_{ij}$.

Step 4. Now we construct the vertex set of G' as the grid points of integer coordinates in Γ, and two vertices are adjacent if and only if their corresponding points in Γ are directly connected on some L_{ij} or S_i for $v_i v_j \in E$ and $1 \le i \le n$, i.e., they are of distance 1 in Γ (see Fig. 1f).

Clearly, G' is a grid graph. Since the maximum degree of G is 3, $m = O(n)$. From Lemma 3.1, $l = O(n^2)$. Thus $|V'| = 3n + 10l + 5m = O(n^2)$, it follows that G' can be constructed in polynomial time. Clearly, G' is a P_2-suspending $5(2l_{ij} + 1)$-subdivision graph of G. By Lemma 2.2 (2) and (3), G has a DS of size at most k if and only if G' has a CSDS of size at most $k + n + 4l + 2m$, as desired. □

Theorem 3.3 *CSDM is NP-complete in supergrid graphs.*

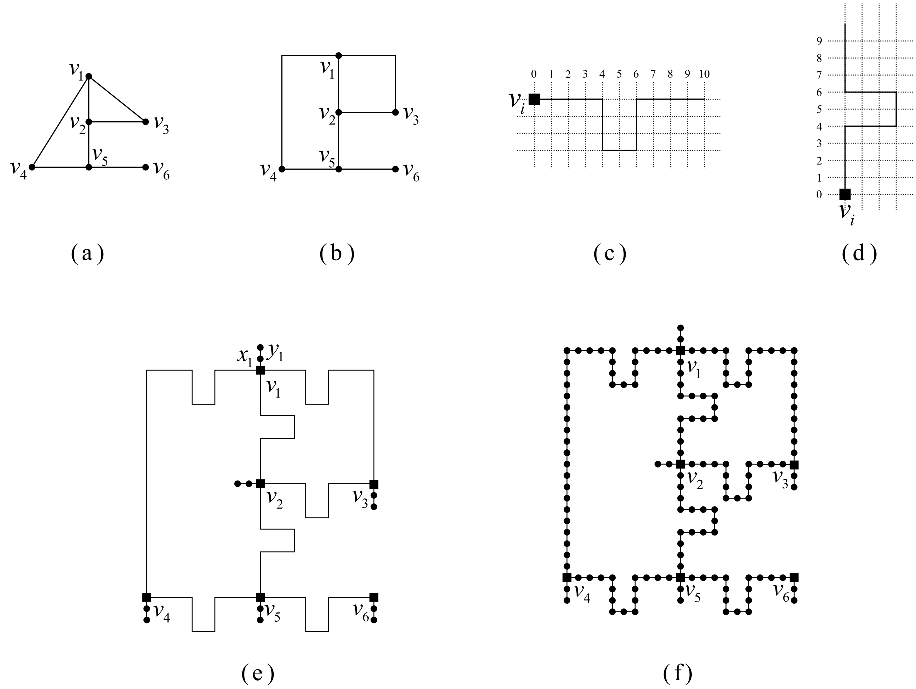

Fig. 1 **a** A simple planar graph G with maximum degree 3. **b** An orthogonal drawing of G. **c** P is horizontal. **d** P is vertical. **e** The drawing Γ. **f** Grid graph G' obtained from G

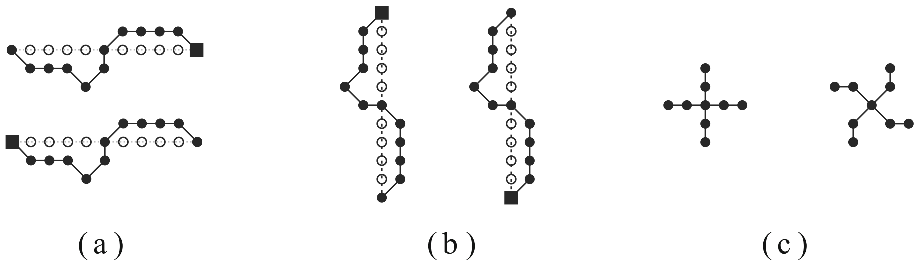

Fig. 2 **a** P is horizontal. **b** P is vertical. **c** The movement of vertices around a bend point

Proof Similar to Theorem 3.2, in what follows we offer a polynomial time reduction from the NP-complete DM-PLA problem. Let $G = (V, E)$ be a simple planar graph with maximum degree 3, where $V = \{v_1, v_2, \ldots, v_n\}$ and $|E| = m$. We construct a supergrid graph $G' = (V', E')$ from G as follows:

Step 1, Step 2. The construction of the two steps is the same as Step 1 and Step 2 of Theorem 3.2.

Step 3. Add vertices on grid points of integer coordinates in Γ'', denoted the resulted drawing still by Γ'', see Fig. 3a. For each of $v_i v_j$-line segment sequence, replace the part P (which is defined in the proof of Theorem 3.2) with the shape as presented in Fig. 2a if P is horizontal, otherwise with the shape as presented in Fig. 2b, see Fig. 3b.

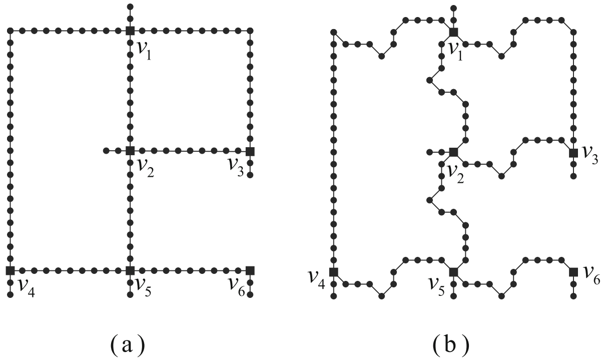

Fig. 3 **a** The drawing Γ''. **b** The drawing obtained after Step 3. **c** Supergrid graph G' obtained from G presented in Fig. 1a

Step 4. For each bend point in Γ'', move up (resp. down, right, left) the two vertices on the left (resp. right, upper, lower) by 1, see Fig. 2c, denoted the resulted drawing by Γ. Note that after processing Step 3, some vertices have been moved into right position.

Step 5. Now we construct the vertex set of G' as the grid points of integer coordinates in Γ, and two vertices u, v are adjacent if and only if their coordinates satisfying $|u_x - v_x| \leq 1$ and $|u_y - v_y| \leq 1$ (see Fig. 3c).

Clearly, G' is a supergrid graph. Since the maximum degree of G is 3, $m = O(n)$. Let $l = \sum_{v_i v_j \in E} l_{ij}$. From Lemma 3.1, $l = O(n^2)$. Thus $|V'| = 3n + 10l = O(n^2)$, it follows that G' can be constructed in polynomial time. Clearly, G' is a P_2-suspending $5(2l_{ij})$-subdivision graph of G. By Lemma 2.2 (2) and (3), G has a DS of size at most k if and only if G' has a CSDS of size at most $k + n + 4l$, as desired. □

Consider two unit squares S_u and S_v. Let (u_x, u_y) (resp. (v_x, v_y)) be coordinate of center of S_u (resp. S_v). Suppose u_x, u_y, v_x, v_y are integers. Then $|u_x - v_x| \leq 1$ and $|u_y - v_y| \leq 1$ if and only if $S_u \cap S_v \neq \emptyset$. Thus we have the following results.

Proposition 3.4 *A graph G is a supergrid graph if and only if G is a unit square graph and the centers of all squares are in integer coordinates.*

Grid graphs are a subclass of unit square graphs (Bertossi and Bonuccelli 1986) and unit disk graphs. By Proposition 3.4, supergrid graphs are a subclass of unit square graphs, but not a subclass of unit disk graphs. By Theorems 3.2 and 3.3, we have the following result.

Corollary 3.5 *CSDM is NP-complete in unit square graphs and unit disk graphs.*

4 APX-hardness

In this section, we show that the MCSDS problem is APX-hard in d-box graphs for any fixed integer $d \geq 2$. Thus the MCSDS problem has no PTAS in d-box graphs for any fixed integer $d \geq 2$.

For the basic optimization terminology we refer to Ausiello et al. (1999). For any NPO optimization problem Π, I_Π is the set of instances of Π, $OPT_\Pi(x)$ is the optimal value of an instance $x \in I_\Pi$, $sol_\Pi(x)$ is the set of feasible solutions for $x \in I_\Pi$, and for each pair (x, y) such that $x \in I_\Pi$ and $y \in sol_\Pi(x)$, $m_\Pi(x, y)$ is the value of a feasible solution y. Let Π_1 and Π_2 be two NPO problems and f be a polynomial time computable function that maps instances of Π_1 to instances of Π_2. Then f is said to be an *L-reduction* from Π_1 to Π_2, if there are positive constants α and β and a polynomial time computable function g such that for every $x \in I_{\Pi_1}$:

(i) $OPT_{\Pi_2}(f(x)) \leq \alpha OPT_{\Pi_1}(x)$,
(ii) for every $y' \in sol_{\Pi_2}(f(x))$, $g(x, y') \in sol_{\Pi_1}(x)$ so that $|OPT_{\Pi_1}(x) - m_{\Pi_1}(x, g(x, y'))| \leq \beta |OPT_{\Pi_2}(f(x)) - m_{\Pi_2}(f(x), y')|$.

To show APX-hardness of a problem it is enough to show that there is an L-reduction from some APX-hard problem.

We first discuss the case of 2-box graphs, that is, rectangle graphs. Given a graph $G = (V, E)$, a *vertex cover* of G is a set $C \subseteq V$ such that each edge $uv \in E$, either $u \in C$ or $v \in C$. The *vertex cover number* $\beta(G)$ is the minimum cardinality among all vertex covers of G. The *minimum vertex cover problem* is a problem of finding a minimum vertex cover in a given graph.

Theorem 4.1 *The MCSDS problem is APX-hard in proper rectangle graphs.*

Proof We offer an L-reduction from the APX-hard problem minimum vertex cover (Alimonti and Kann 2000) in a graph $G = (\{v_1, v_2, \ldots, v_n\}, E)$ with maximum degree at most 3 to the MCSDS problem in a proper rectangle graph G'. Let $|E| = m$. For each edge $v_i v_j \in E$, suppose $i < j$ and denote $e_{ij} = v_i v_j$. Using the similar method as in Erlebach and van Leeuwen (2008); Wang et al. (2023a), we construct a proper rectangle graph G' as follows:

Step 1. For each vertex v_i, construct a thin horizontal rectangle r_i^h, a thin vertical rectangle r_i^v and three little rectangles a_i, b_i, c_i. Place $r_1^h, \ldots, r_n^h$ from top to bottom that do not intersect each other such that their left boundaries are on a vertical line l. Place $r_1^v, \ldots, r_n^v$ from left to right that do not intersect each other such that: (1) the left boundary of r_1^v is on the vertical line l, (2) the top boundary of r_i^v is between the bottom boundary of r_i^h and top boundary of r_{i+1}^h for each $1 \leq i \leq n$ (r_{n+1}^h can be regarded as a virtual rectangle below the r_n^h). For any $1 \leq i \leq n$, a_i, b_i, c_i are placed such that: (1) b_i intersects r_i^v, r_i^h, (2) a_i (resp. c_i) intersects b_i, c_i, r_i^h (resp. a_i, b_i, r_i^v).

Step 2. For any r_i^h (resp. r_i^v), construct two same little squares p_i^h, q_i^h (resp. p_i^v, q_i^v) such that p_i^h (resp. p_i^v) only intersects r_i^h and q_i^h (resp. r_i^v and q_i^v), q_i^h (resp. q_i^v) only intersects p_i^h (resp. p_i^v).

Step 3. For each r_i^v, there is an intersection region S_{ij} with r_j^h, where $1 \leq i < j \leq n$. For each $e_{ij} \in E$, construct three same little squares x_{ij}, y_{ij}, z_{ij} such that z_{ij} only intersects y_{ij}, y_{ij} only intersects z_{ij} and x_{ij}, and x_{ij} intersects y_{ij} and S_{ij}.

Step 4. Adjusting the size and position of rectangles appropriately will result in rectangles not containing each other. Figure 4b is an example of the construction.

Given a minimum vertex cover C of G. Let $S = \{r_i^h, r_i^v : v_i \in C\} \cup \{b_i : v_i \notin C\} \cup \{y_{ij} : e_{ij} \in E\} \cup \{p_i^h, p_i^v : 1 \leq i \leq n\}$. Next, we show that S is a CSDS of G'. Clearly, S is a DS of G'. For each $1 \leq i \leq n$, if $r_i^h, r_i^v \in S$, then b_i replaces r_i^h, r_i^v. Otherwise $b_i \in S$, then a_i replaces b_i. Let $H_i = \{a_i, b_i, c_i, r_i^h, r_i^v\}$, where $1 \leq i \leq n$. For each $1 \leq i \leq n$, since $S \cap H_i$ is a DS of $G'[H_i]$, q_i^h replaces p_i^h and q_i^v replaces p_i^v. Since C is a vertex cover of G, there is $v_i \in C$ or $v_j \in C$ for each $e_{ij} \in E$, that is, $r_i^h, r_i^v \in S$ or $r_j^h, r_j^v \in S$. Thus for each y_{ij},

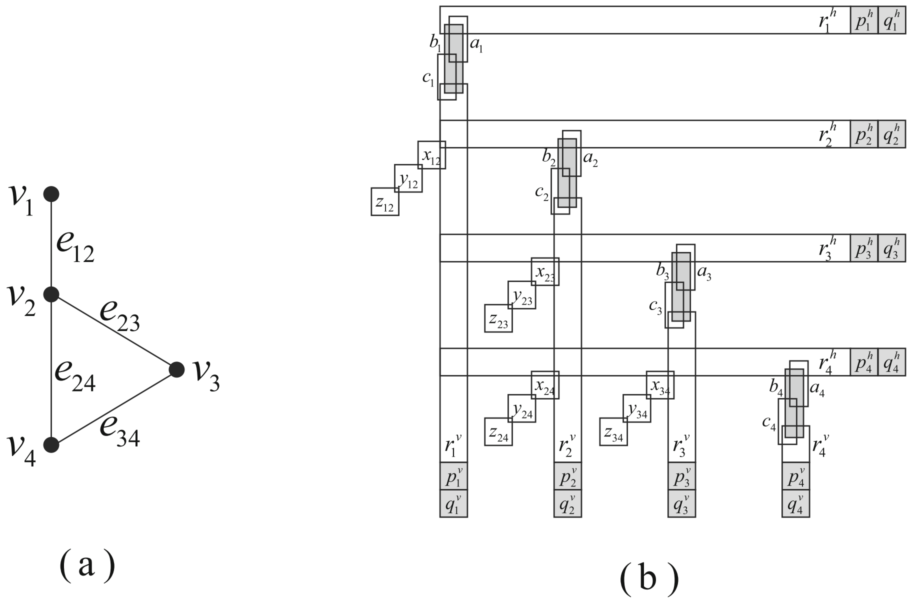

Fig. 4 **a** A graph G with maximum degree 3. **b** A proper rectangle graph G' obtained from G

z_{ij} replaces y_{ij}. Since the maximum degree of G is 3, $\beta(G) \geq n/4$ and $m \leq 3n/2$. Hence, $\gamma_{cs}(G') \leq |S| = |C| + 3n + m = \beta(G) + 3n + m \leq 19\beta(G)$.

Given a CSDS S of G'. We will construct a CSDS S' of G' from S satisfying the following conditions: (1) $|S'| \leq |S|$; (2) for each $e_{ij} \in E$, $z_{ij}, x_{ij} \notin S'$; (3) for each $1 \leq i \leq n$, $q_i^h, q_i^v \notin S'$; (4) for each $1 \leq i \leq n$, $S' \cap H_i = \{r_i^h, r_i^v\}$ or $|S' \cap H_i| = 1$.

By Lemma 2.1, S can be transformed in polynomial time to a CSDS S_1 of G' such that $|S_1| \leq |S|$, $z_{ij} \notin S_1$ for each $e_{ij} \in E$ and $q_i^h, q_i^v \notin S_1$ for each $1 \leq i \leq n$. Then $y_{ij} \in S_1$ for each $e_{ij} \in E$ and $p_i^h, p_i^v \in S_1$ for each $1 \leq i \leq n$. For each $x_{ij} \in S_1$, $r_i^v \notin S_1$ or $r_j^h \notin S_1$, without loss of generality, suppose $r_j^h \notin S_1$, then let $S_2 = (S_1 \backslash \{x_{ij}\}) \cup \{r_j^h\}$. For each $1 \leq i \leq n$, if $|S_2 \cap H_i| \geq 2$, then let $S' = (S_2 \backslash H_i) \cup \{r_i^h, r_i^v\}$. Obviously, S' is a CSDS of G' satisfying the conditions (1), (2), (3) and (4).

Let $C = \{v_i : r_i^v, r_i^h \in S'\}$. For each $e_{ij} \in E$, z_{ij} is unique replaceable vertex of y_{ij}, that is, $(S' \backslash \{y_{ij}\}) \cup \{z_{ij}\}$ is a DS of G'. Since $x_{ij} \notin S'$, $r_i^v \in S'$ or $r_j^h \in S'$, without loss of generality, suppose $r_j^h \in S'$. Since c_j needs to be dominated by S', $|(S' \cap H_j) \backslash \{r_j^h\}| \geq 1$. Thus $|S' \cap H_j| \geq 2$, it follows that $S' \cap H_j = \{r_j^h, r_j^v\}$ by condition (4). Then $v_j \in C$. Hence, C is a vertex cover of G. Due to conditions (1), (2), (3) and (4), $|S| \geq |S'| = |C| + 3n + m$. If S is a minimum CSDS of G', then $\gamma_{cs}(G') = |S| \geq |C| + 3n + m \geq \beta(G) + 3n + m$. Hence, $\gamma_{cs}(G') = \beta(G) + 3n + m$. Thus $|S| - \gamma_{cs}(G') \geq |C| - \beta(G)$. □

By Theorem 4.1 and proper rectangle graphs are a subclass of rectangle graphs, we have the following result.

Corollary 4.2 *The MCSDS problem is APX-hard in rectangle graphs.*

Now we consider the case of $d \geq 3$.

Lemma 4.3 (Wang et al. 2023c) *Given a graph $G = (\{v_1, v_2, \ldots, v_n\}, E)$ and an integer $s_{ij} \geq 1$ for each edge $v_i v_j \in E$. Let G' be a P_2-suspending $5s_{ij}$-subdivision graph of G. Then for any fixed integer $d \geq 3$, the graph G' can be realized as a d-box graph. Moreover, such realization can be done in time polynomial in $3|V| + \sum_{v_i v_j \in E} s_{ij}$.*

Theorem 4.4 *The MCSDS problem is APX-hard in d-box graphs, where $d \geq 3$.*

Proof We offer an L-reduction from the APX-hard problem minimum domination (Alimonti and Kann 2000) in a graph $G = (V, E)$ with maximum degree at most 3 to the MCSDS problem in a d-box graph G'. Let $|V| = n$ and $|E| = m$. Let G' be a P_2-suspending 5-subdivision graph of G. By Lemma 4.3, G' is a d-box graph.

Given a minimum DS D of G. By Lemma 2.2 (2), D can be transformed in polynomial time to a CSDS S of G' such that $|S| \leq |D| + n + 2m$. Since the maximum degree of G is 3, $\gamma(G) \geq n/4$ and $m \leq 3n/2$. Hence, $\gamma_{cs}(G') \leq |S| \leq |D| + n + 2m = \gamma(G) + n + 2m \leq 17\gamma(G)$.

Given a CSDS S of G'. By Lemma 2.2 (3), S can be transformed in polynomial time to a DS D of G such that $|S| \geq |D| + n + 2m$. By Lemma 2.2 (1), $\gamma_{cs}(G') = \gamma(G) + n + 2m$. Thus $|S| - \gamma_{cs}(G') \geq |D| - \gamma(G)$. □

5 Constant-approximation algorithm

In this section, we give an $O(n + m)$ time $2(t-1)$-approximation algorithm for the MCSDS problem in $K_{1,t}$-free graphs with n vertices and m edges, where $t \geq 3$ is an integer. The pseudo code is presented in Algorithm 1. Further, we discuss the MCSDS problem in several geometric intersection graphs which are $K_{1,t}$-free for some integer $t \geq 3$.

For a graph $G = (V, E)$, an *independent set* of G is a set $I \subseteq V$ such that any two vertices in I are not adjacent. An independent set I is referred to as a *maximal independent set* if for any vertex $v \in V \backslash I$, $I \cup \{v\}$ is not an independent set. The *independence number* $\alpha(G)$ of G is the maximum cardinality among all independent sets of G.

Algorithm 1

Input: A general graph G.
Output: A CSDS I of G.
$A = B = \emptyset$;
Compute a maximal independent set I_1 of G such that all leaves of G are in I_1.
for each $u \in I_1$
  pick one vertex $u' \in N_G(u)$, $A \leftarrow A \cup \{u'\}$;
end for
Compute a maximal independent set I_2 of $G - I_1 - A$;
for each $u \in I_2$
  if $N_G(u) \subseteq I_1$, **then** $I_2 \leftarrow I_2 \backslash \{u\}$, $B \leftarrow B \cup \{u\}$;
  end if
end for
Return $I = I_1 \cup I_2$.

Lemma 5.1 *I is a CSDS of G.*

Proof Since I_1 is a maximal independent set of G, I is a DS of G. For each $u \in I_1$, there is unique $u' \in A \cap N_G(u)$. In what follows, we show that $(I\backslash\{u\}) \cup \{u'\}$ is a DS of G. We only need to consider any vertex $x \in N_G(u)$ and $x \notin I \cup \{u'\}$. If $x \in A$, then there is $v \in I_1\backslash\{u\} \subseteq (I\backslash\{u\}) \cup \{u'\}$ such that $x \in N_G(v)$, as desired. If $x \in B$, then $N_G(x) \subseteq I_1$. Since $x \notin I_1$, x is not a leaf, it follows that x has another neighbor $v \neq u$ in I_1, as desired. Now we assume that $x \notin A \cup B$. Let $I_2^0 = I_2 \cup B$. Then I_2^0 is the maximal independent set of $G - I_1 - A$ computed in Step 6. If $N_G(x) \cap I_2 = \emptyset$, then $I_2 \cup \{x\}$ is independent in $G - (I_1 \cup A)$. Since all neighbors of each vertex in B are in I_1 and $x \notin I_1$, x has no neighbor in B. Thus $I_2^0 \cup \{x\}$ is independent, which is a contradiction with I_2^0 is a maximal independent set of $G - I_1 - A$. Then there is a vertex $v \in N_G(x) \cap I_2$, as desired.

Now we consider each vertex $u \in I_2$. If $N_G(u) \subseteq I$, then $N_G(u) \subseteq I_1$ by I_2 is independent. Thus $u \in B$, which is a contradiction with $u \in I_2$. Then there is a vertex $v \in N_G(u)$ such that $v \notin I$. Now we show that $(I\backslash\{u\}) \cup \{v\}$ is a DS of G. We only need to consider any $x \in N_G(u)$ and $x \notin I \cup \{v\}$. If $N_G(x) \cap I_1 = \emptyset$, then $I_1 \cup \{x\}$ is independent, which is a contradiction with I_1 is a maximal independent set of G. There is a vertex $y \in N_G(x) \cap I_1$. Then $y \in (I\backslash\{u\}) \cup \{v\}$, as desired. □

Note that Algorithm 1 produces a CSDS for general graphs. In the following we show that Algorithm 1 is $2(t-1)$-approximation for $K_{1,t}$-free graphs, where $t \geq 3$. Given a graph $G = (V, E)$ and a set $S \subseteq V$, an *S-external private neighbor* of a vertex $v \in S$ is a vertex $u \in V\backslash S$ such that $N_G(u) \cap S = \{v\}$. The *S-external private neighborhood* of $v \in S$, denoted by $epn(v, S)$, is the set of all S-external private neighbors of v. If a vertex $v \in V$ is adjacent to all vertices in $V\backslash\{v\}$, then v is called a *universal vertex*.

Lemma 5.2 *Let $G = (V, E)$ be a $K_{1,t}$-free graph, where $t \geq 3$ is an integer. Then $\alpha(G) \leq (t-1)\gamma_{cs}(G)$. Moreover, the bound is tight.*

Proof Let $S = \{v_1, \ldots, v_k, u_1, \ldots, u_s\}$ be a minimum CSDS of G for some integers $k, s \geq 0$, where v_i, u_j satisfy $epn(v_i, S) \neq \emptyset$, $1 \leq i \leq k$ and $epn(u_j, S) = \emptyset$, $1 \leq j \leq s$. Let $C = V\backslash(S \cup epn(v_1, S) \cup \cdots \cup epn(v_k, S))$. Then each vertex in C has at least 2 neighbors in S. Let $C_1 = \{w \in C : N_G(w) \cap \{u_1, \ldots, u_s\} = \emptyset\}$, $C_2 = \{w \in C : N_G(w) \cap \{v_1, \ldots, v_k\} = \emptyset\}$ and $C_3 = C\backslash(C_1 \cup C_2)$.

Let I be a maximum independent set of G, and $I_i = I \cap C_i$ for $i = 1, 2, 3$. Without loss of generality, suppose $I \cap \{v_1, \ldots, v_k\} = \{v_1, \ldots, v_a\}$ for some integer $0 \leq a \leq k$ and $I \cap \{u_1, \ldots, u_s\} = \{u_1, \ldots, u_h\}$ for some integer $0 \leq h \leq s$. Then for any $1 \leq i \leq a$, $epn(v_i, S) \cap I = \emptyset$. For any $a + 1 \leq i \leq k$, suppose $|epn(v_i, S) \cap I| = t_i$. Since G is $K_{1,t}$-free, $0 \leq t_i \leq t - 1$.

Consider a bipartite graph H_1 with bipartition $\{v_{a+1}, \ldots, v_k\}$ and $I_1 \cup I_3$, and $E(H_1) = \{v_i w \in E : a + 1 \leq i \leq k, w \in I_1 \cup I_3\}$. For each $a + 1 \leq i \leq k$, since G is $K_{1,t}$-free, v_i has at most $(t - 1 - t_i)$ neighbors in $I_1 \cup I_3$. Then

$$|E(H_1)| \leq t - 1 - t_{a+1} + \cdots + t - 1 - t_k = (k - a)(t - 1) - (t_{a+1} + \cdots + t_k). \quad (5.1)$$

Note that each vertex in $I_1 \cup I_3$ has no neighbor in $\{v_1, \ldots, v_a\}$, $|N_G(w) \cap \{v_1, \ldots, v_k\}| \geq 2$ (resp. $|N_G(w) \cap \{v_1, \ldots, v_k\}| \geq 1$) for each $w \in I_1$ (resp. $w \in I_3$) by the definition of C_1 (resp. C_3). Thus each vertex in I_1 (resp. I_3) has at least 2 neighbors (resp. one neighbor) in $\{v_{a+1}, \ldots, v_k\}$ (see Fig. 5), it follows that

$$|E(H_1)| \geq 2|I_1| + |I_3|. \quad (5.2)$$

Combining inequalities (5.1) with (5.2),

$$|I_1| \leq \frac{(k-a)(t-1)}{2} - \frac{(t_{a+1} + \cdots + t_k)}{2} - \frac{|I_3|}{2}. \quad (5.3)$$

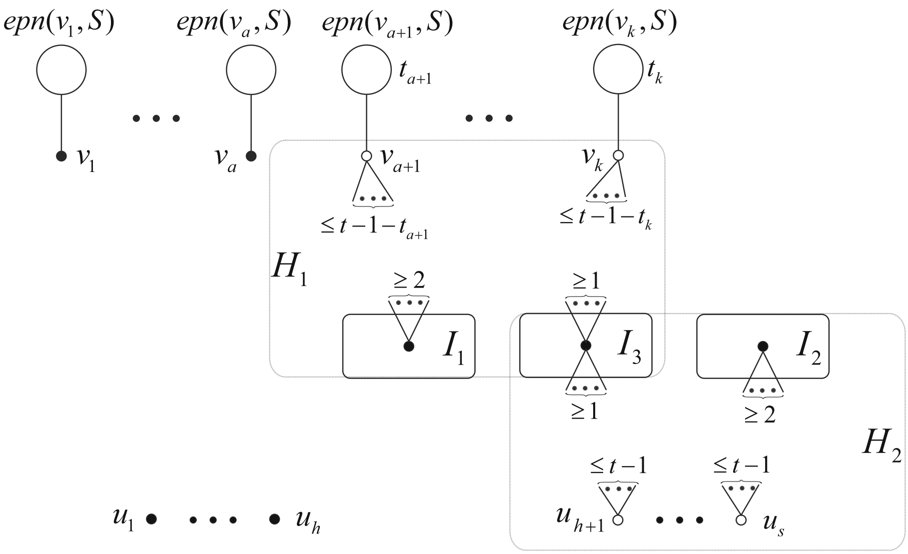

Fig. 5 Illustration for Lemma 5.2. The black vertices represent the vertices in I, the large cycle above v_i represents the S-external private neighborhood of v_i for each $1 \leq i \leq k$ and the number t_i above v_i represents that there are t_i vertices in $epn(v_i, S) \cap I$ for each $a+1 \leq i \leq k$. For each $a+1 \leq i \leq k$, v_i has at most $(t-1-t_i)$ neighbors in $I_1 \cup I_3$. For each $h+1 \leq j \leq s$, u_j has at most $(t-1)$ neighbors in $I_2 \cup I_3$. Each vertex in I_1 has at least 2 neighbors in $\{v_{a+1}, \ldots, v_k\}$. Each vertex in I_2 has at least 2 neighbors in $\{u_{h+1}, \ldots, u_s\}$. Each vertex in I_3 has at least one neighbor in $\{v_{a+1}, \ldots, v_k\}$ and $\{u_{h+1}, \ldots, u_s\}$, respectively. Bipartite graph H_1 with bipartition $\{v_{a+1}, \ldots, v_k\}$ and $I_1 \cup I_3$, bipartite graph H_2 with bipartition $\{u_{h+1}, \ldots, u_s\}$ and $I_2 \cup I_3$

Similarly, consider a bipartite graph H_2 with bipartition $\{u_{h+1}, \ldots, u_s\}$ and $I_2 \cup I_3$, and $E(H_2) = \{u_j w \in E : h+1 \leq j \leq s, w \in I_2 \cup I_3\}$. Each vertex in $\{u_{h+1}, \ldots, u_s\}$ has at most $(t-1)$ neighbors in $I_2 \cup I_3$. Then

$$|E(H_2)| \leq (t-1)(s-h). \tag{5.4}$$

Note that each vertex in $I_2 \cup I_3$ has no neighbor in $\{u_1, \ldots, u_h\}$, $|N_G(w) \cap \{u_1, \ldots, u_s\}| \geq 2$ (resp. $|N_G(w) \cap \{u_1, \ldots, u_s\}| \geq 1$) for each $w \in I_2$ (resp. $w \in I_3$) by the definition of C_2 (resp. C_3). Thus each vertex in I_2 (resp. I_3) has at least 2 neighbors (resp. one neighbor) in $\{u_{h+1}, \ldots, u_s\}$ (see Fig. 5). Then

$$|E(H_2)| \geq 2|I_2| + |I_3|. \tag{5.5}$$

Combining inequalities (5.4) with (5.5),

$$|I_2| \leq \frac{(t-1)(s-h)}{2} - \frac{|I_3|}{2}. \tag{5.6}$$

By the inequalities (5.3), (5.6) and $t \geq 3$, we have the following

$$\begin{aligned}
\alpha(G) &= |I| = a + h + (t_{a+1} + \cdots + t_k) + |I_1| + |I_2| + |I_3| \\
&\leq a + h + (t_{a+1} + \cdots + t_k) + \frac{(k-a)(t-1)}{2} - \frac{(t_{a+1} + \cdots + t_k)}{2} - \frac{|I_3|}{2} \\
&\quad + \frac{(t-1)(s-h)}{2} - \frac{|I_3|}{2} + |I_3|
\end{aligned}$$

$$
\begin{aligned}
&= a + h + \frac{(t_{a+1} + \cdots + t_k)}{2} + \frac{(k-a)(t-1)}{2} + \frac{(t-1)(s-h)}{2} \\
&\le a + h + \frac{(k-a)(t-1)}{2} + \frac{(k-a)(t-1)}{2} + \frac{(t-1)(s-h)}{2} \\
&= (t-1)k + \frac{t-1}{2}s - (t-2)a - \frac{t-3}{2}h \\
&\le (t-1)k + \frac{t-1}{2}s \\
&\le (t-1)k + (t-1)s \\
&= (t-1)\gamma_{cs}(G).
\end{aligned}
$$

Let G be a $K_{1,t}$-free graph whose vertex set can be partition into two subsets I and C, where I is an independent set and $|I| = t-1$, C contains two universal vertices u and v. Then $\alpha(G) = (t-1)\gamma_{cs}(G)$. In fact, I is a maximum independent set of G. Otherwise there is an independent set I' of G with $|I'| > |I|$. Since u is a universal vertex, u is adjacent to each vertex in I', which is a contradiction with G is $K_{1,t}$-free. Thus $\alpha(G) = |I| = t-1$. The set $S = \{u\}$ is a CSDS of G. In fact, S is a DS of G and $(S\backslash\{u\}) \cup \{v\} = \{v\}$ is a DS of G by v is a universal vertex. Then $\gamma_{cs}(G) = 1$. Thus $\alpha(G) = (t-1)\gamma_{cs}(G)$. □

Theorem 5.3 *Given a $K_{1,t}$-free graph G with n vertices and m edges, where $t \ge 3$ is an integer, Algorithm 1 produces a CSDS of G with size at most $2(t-1)\gamma_{cs}(G)$ in $O(n+m)$ time.*

Proof By Lemma 5.1, Algorithm 1 produces a CSDS I of G. Clearly, I_1 and I_2 are independent sets in G. By Lemma 5.2, $|I| = |I_1| + |I_2| \le 2\alpha(G) \le 2(t-1)\gamma_{cs}(G)$. Since a maximal independent set can be obtained in $O(n+m)$ time, Algorithm 1 runs in $O(n+m)$ time. □

Now we present several geometric intersection graphs which are $K_{1,t}$-free for some integer $t \ge 3$.

Proposition 5.4 (Hale 1980) *Unit disk graphs are $K_{1,6}$-free.*

Proposition 5.5 (Butenko et al. 2011) *Unit ball graphs are $K_{1,13}$-free.*

Proposition 5.6 (Wang et al. 2023a) *Given a unit-height rectangle representation, then the corresponding unit-height rectangle graph is $K_{1,2\lceil w\rceil+3}$-free, where $w \ge 1$ is the maximum width of all rectangles in the representation. Further, unit square graphs are $K_{1,5}$-free.*

By Theorem 5.3 and Propositions 5.4, 5.5, 5.6, we have the following results.

Theorem 5.7 *Algorithm 1 is an $O(n+m)$ time 8-approximation (resp. 10-approximation, 24-approximation) algorithm for the MCSDS problem in unit square graphs (resp. unit disk graphs, unit ball graphs) with n vertices and m edges for the case that a unit square representation (resp. unit disk representation, unit ball representation) is not given.*

Theorem 5.8 *Algorithm 1 is an $O(n+m)$ time $(4\lceil w\rceil + 4)$-approximation algorithm for the MCSDS problem in unit-height rectangle graphs with n vertices and m edges whose unit-height rectangle representation is given, where $w \ge 1$ is the maximum width of all rectangles in the representation.*

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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